

MOTIVATION & PURPOSE

The BC housing market is one of Canada's most challenging real estate environments, with property prices spanning from modest rural homes to multi-million-dollar Vancouver estates. Buyers, sellers, investors, and policy-makers all need accurate, data-driven price estimates — yet existing tools often lack transparency or regional nuance. **Our Goal:** Build a supervised machine learning system that predicts housing prices in British Columbia from property features, providing accurate estimates that can assist home buyers and real-estate professionals.

Key Questions Addressed:

- Which property features (size, location, type) matter most for BC pricing?
- Can ensemble methods outperform linear baselines on high-dimensional real-estate data?
- How accurately can we predict prices across diverse BC regions?

DATASET & PROJECT OVERVIEW

21,977

Property Records

9

Raw Features

243

Engineered Features

Features include: City, Property Type (categorical → one-hot encoded); Latitude, Longitude, Bedrooms, Bathrooms, Square Footage (numerical → standardized)

The preprocessing pipeline applies one-hot encoding on categorical variables and StandardScaler on numerical ones, expanding the dataset from 9 raw features to 243 engineered features. An 80/20 train-test split is used for all model evaluations.

METHODS & TECHNOLOGIES

Six Regression Models Evaluated:

- Linear Regression** Baseline linear model.
- Ridge Regression** L2 regularisation; α tuned via GridSearchCV.
- Lasso Regression** L1 regularisation; feature-selecting.
- Decision Tree** Non-linear tree; depth and split params tuned.
- K-Nearest Neighbours** Distance-weighted kNN; $k \in \{3, 5, 7, 9\}$ tested.
- Random Forest** Ensemble of 100–200 trees; best performer.

All models share the same preprocessing pipeline and are tuned via GridSearchCV with cross-validation. Results are compared on MAE, RMSE, and R^2 .

AI PIPELINE

①

Raw Data

21,977 BC property listings

②

Preprocessing

Encode, scale, train/test split

③

Feature Eng.

243 features from 9 raw

④

Model Training

6 models + GridSearchCV

⑤

Evaluation

MAE, RMSE, R^2 metrics

HYPERPARAMETER TUNING

GridSearchCV with cross-validation was used to tune all five non-baseline models. The best hyperparameters by lowest RMSE are saved as .pkl pipelines for reproducible evaluation.

Model	Key Parameters Tuned	Best Config (GridSearchCV)
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf	200 trees, max_depth=20, split=2, leaf=2
Decision Tree	max_depth, min_samples_split, min_samples_leaf	max_depth=10, split=10, leaf=4
KNN	n_neighbors, weights, p (distance metric)	k=9, distance weights, Manhattan (p=1)
Ridge	$\alpha \in \{0.1, 1.0, 10.0, 50.0, 100.0\}$	$\alpha=1.0$
Lasso	$\alpha \in \{0.0001-1.0\}$, max_iter=5000	$\alpha=1.0$, max_iter=5000

REFERENCES

- Scikit-learn developers. (2024). sklearn.ensemble.RandomForestRegressor. scikit-learn.org
- Government of BC. (2024). BC Assessment Data Catalogue. catalogue.data.gov.bc.ca
- Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825–2830.
- Repository: github.com/ammарprasla/BC-Housing-Price-Predictor

RESULTS – BEST MODEL: RANDOM FOREST

\$228,287

Mean Absolute Error

\$405,301

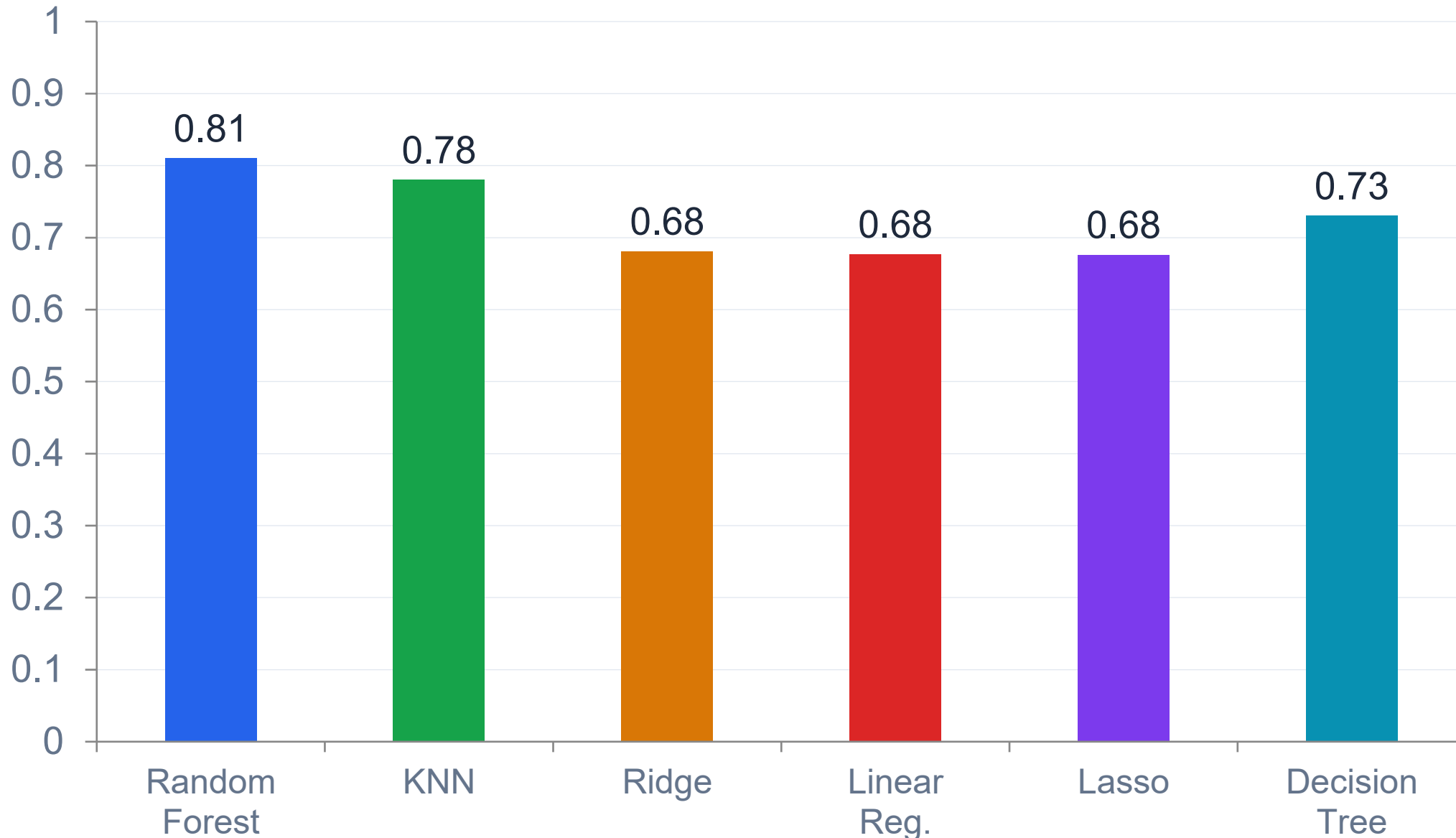
Root Mean Sq. Error

0.8143

 R^2 Score

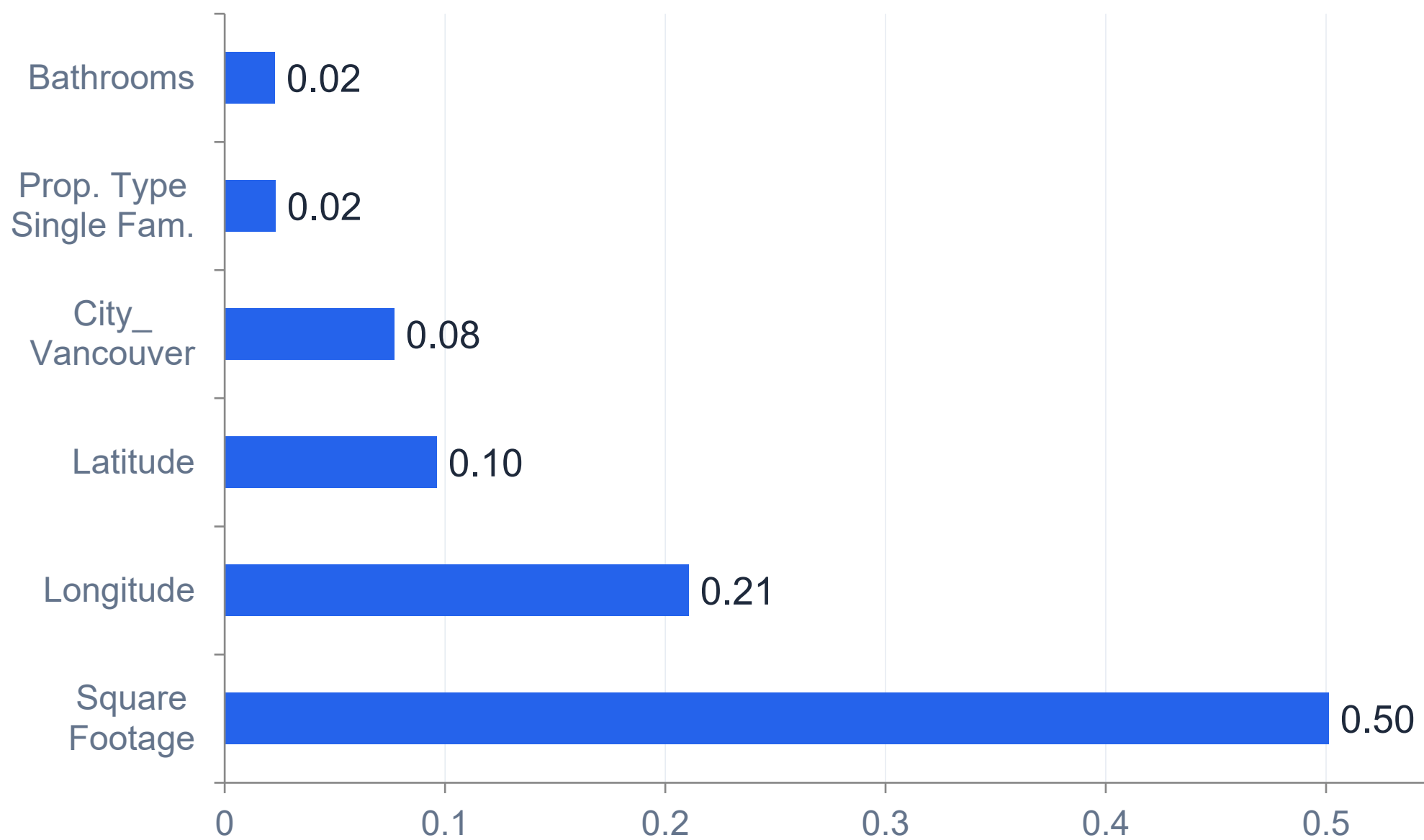
Random Forest achieved the best performance across all metrics — an R^2 of 0.81 means the model explains over 81% of the variance in housing prices. The relatively high dollar values for MAE/RMSE reflect the wide price range in BC real estate (\$100k–\$5M+).

MODEL PERFORMANCE COMPARISON

 R^2 Score by Model (↑ Higher is Better)

RANDOM FOREST – TOP FEATURE IMPORTANCES

Mean Decrease in Impurity



KEY FINDINGS & CONCLUSION

Square Footage Dominates (50.1% importance): Physical size is the single strongest predictor of housing price in BC.

Location is Critical (Longitude + Latitude + City_Vancouver $\approx$ 38% combined): Geography, especially proximity to the Greater Vancouver area, is the second most powerful signal.

Ensemble Models Win Random Forest outperformed all other approaches. Its R^2 of 0.81 represents a substantial improvement over linear methods ($R^2 \approx 0.68$), confirming that non-linear interactions between features matter in real-estate pricing.

High Dimensionality Challenges One-hot encoding city names creates 230+ columns, causing KNN to suffer from the curse of dimensionality and Lasso to struggle with convergence.

Future Work:

- Incorporate time-series data and market trends
- Add features such as proximity to transit and school ratings
- Explore XGBoost and neural network approaches
- Deploy as a web tool for public price estimation